Anakala Sashi Kiran

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	8.01
FIITJEE Jr. College	Class XII (Telangana State Board of Intermediate Education)	2023	9.33%
Global Edge School Kukatpally	Class X (CBSE)	2021	9.25%

WORK EXPERIENCE

Samsung R&D India (Full-time Intern)

- Engineered and optimized a SLAM (Simultaneous Localization and Mapping) component for **flagship AR Glasses**.
- Conducted a literature review of **20+ research papers** to select baseline algorithms and tools for spatial tracking.
- Designed and experimented with **novel tracking metrics**, increasing performance evaluation accuracy.

IIT Hyderabad (Undergrad Research Contributor - Distributed Systems)

- Conceptualized and implemented **FLAKE** (Federated Learning with Asynchronous non-IID data and Knowledge Exchange) to handle heterogeneous client data.
- Achieved a near **30%** improvement in accuracy for ResNet-20 on the CIFAR dataset over the previous SOTA on Layer Sharing.
- Integrated Knowledge Distillation techniques into asynchronous FL systems to improve **convergence** and model **robustness**.

PROJECTS

LendDAO: Multi-Token DeFi Protocol & Governance

- Programmed a **multi-token lending pool** on Aptos, allowing borrowers to withdraw against overcollateralized positions
- Coded an algorithmic **interest rate calculator** and **liquidator** module, triggers when loan-to-value ratios cross safety thresholds.
- Structured an on-chain treasury to collect the interest rate spread, enabling token holders to distribute dividends, subsidize lending pools.
- Built a **Rust CLI** leveraging the Aptos Rust SDK for administrative interaction and contract execution, and integrated Petra Wallet with a **React.js** frontend interface.

2D Level-Up Shootout: Kingdom

- Developed a **2D sandbox-style** game in C++ under **3000 lines** of code utilizing **Raylib**, implementing a tiled coordinates
- Structured the codebase around dedicated managers, building **tile polymorphism** using a base Tile class, designing an entity inheritance hierarchy derived from a base Entity class, and an inventory system to manage crafting, building, and harvesting mechanics.
- **Playtested** with several real users, achieving a playtime of over **5 hours** among the most dedicated players

Peer-to-Peer (P2P) Mesh Chat Network

- Developed an **autonomous P2P messaging network** implementing a custom Distance-Vector routing protocol for multi-hop message forwarding (A -> B -> C).
- Designed a presence protocol that **isolates user identities** from UUID session IDs, minimizing exposure of the social graph.
- Programmed **periodic Distance-Vector advertisements** to propagate shortest path metrics to neighbors.

QUIRC: Quantum IR Compiler

- Built a compiler frontend using Flex and Bison to compile a **hybrid classical-quantum language** (QuirC) supporting qubit types, classical control loops, structures, and quantum operations.
- Defined a **custom MLIR dialect** using LLVM TableGen (Dialect.td, Types.td, Ops.td), implementing dialect operations
- Translated AST nodes into compiler intermediate representations, establishing a **pipeline for optimization passes**.

SKILLS

- **Languages:** C, C++, CUDA, Python, HTML, CSS, SQL, RISC-V
- **Tools:** VS Code, Jupyter Notebook, Vim, NeoVim, Antigravity, NotebookLM, TMUX, Bash, Ubuntu, GCC, G++, Numpy, Tensorflow, Raylib, Github, Latex, Ripes

RELEVANT COURSES

- **Mathematics:** Discrete Mathematics for Computer Science, Probability and Random Variables, Transform Techniques, Introduction to Statistics, Linear Algebra, Theory of Computation, Algebra: Groups and Rings

- **Computer Science:** Database Management Systems, Operating Systems, Computer Architecture, Data Structures and Algorithms, Compilers, Software Engineering, Computer Networks, Distributed Computing, Machine Learning, Deep Learning, GPU Programming, Practical Malware Analysis

- **Coursera (ML Specialization):** Supervised ML - Regression and Classification, Advanced ML Algorithms

EXTRACURRICULAR

- **Extracurriculars:** Badminton, Swimming, Tennis, Chess, Reading, Writing, learning Japanese, Piano, Crosswords
- **Miscellaneous:** Spell Bee State Finalist, 3-time MUN participant, Actor&Scriptwriter for a Japanese-language short skit, Author of 3 blogs + a Substack

POSITIONS OF RESPONSIBILITY

- Teaching Assistant for **Compilers I**, and **Fundamentals of Machine Learning**, Aksharmala volunteer (current)
- International Relationships Cell, Books & Movies Club, and Vibes Music Club Member, Farewell 2025 **Core Organising Team**, National Service Scheme (NSS) Volunteering Member

SCHOLASTIC ACHIEVEMENTS

- KVPY Scholar (Inspire fellow), ASSET Scholar (**99.9%ile** in 2/3 subjects)
- SOF School **Rank 1**, Zonal **Rank 13**, International **Rank 15**, CCMB YIP 2019 Scholar